

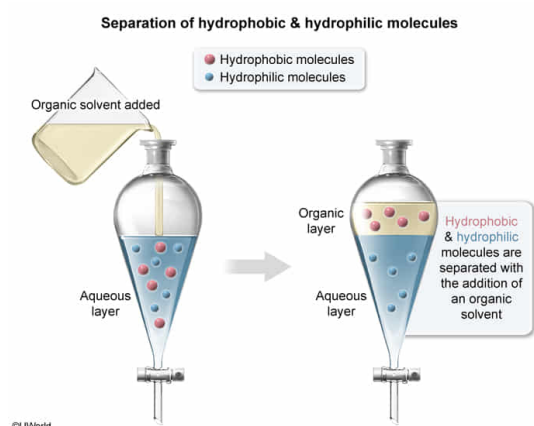
The solutes in an aqueous reaction mixture are separated by extraction. The function of the nonpolar organic solvent is to:

- ☐ A. transfer hydrophilic molecules to the organic layer. [6%]
- ☐ B. combine the components of a reaction mixture. [1%]
- ☐ C. increase the volume of the solvent.
- ✓ ☒ D. remove hydrophobic molecules from the aqueous layer. [91%]

Correct

92% answered correctly

Explanation:



Extraction is a technique used to **separate molecules** based on their **solubility**. It relies on the principle that "**like dissolves like**", meaning that molecules will dissolve in a **solvent** of the **same polarity** (ie, nonpolar molecules will dissolve in nonpolar solvents and polar molecules will dissolve in polar solvents).

If the initial reaction mixture contains hydrophobic and hydrophilic molecules in an aqueous (polar) solution, a nonpolar organic solvent is added to the aqueous solution in a separatory funnel for an extraction. The separatory funnel is shaken, and the funnel is vented to release pressure. The organic and aqueous layers are immiscible (meaning they do not mix) so after shaking, the layers separate based on their densities. Hydrophilic ("water-loving") molecules are polar and will remain in the aqueous layer. Hydrophobic ("water-fearing") molecules are nonpolar and will dissolve better in the organic layer. Therefore, a **nonpolar organic solvent** is used in an extraction to remove the **hydrophobic molecules** from the polar aqueous layer while the **hydrophilic molecules** remain in the **aqueous layer**.

(Choice A) Hydrophilic molecules are polar and will stay in the polar aqueous layer rather than go into the nonpolar organic layer.

(Choice B) The components of the reaction mixture are already combined before the extraction. An extraction is done to *separate* molecules rather than combine them.

(Choice C) Addition of an organic solvent to the aqueous layer will increase the volume, but this is not the purpose of the organic solvent.

Educational objective:

Extraction is a technique used to separate molecules based on their solubility in aqueous (polar) or organic (nonpolar) phases. The mixture components will dissolve in a solvent of the same polarity ("like dissolves like"). Hydrophobic (nonpolar) molecules will dissolve in a nonpolar organic solvent, and hydrophilic (polar) molecules will dissolve in an aqueous solution.

Time spent:0 sec

Subject:
Organic Chemistry

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System:
Separation Techniques, Spectroscopy, and Analytical Methods